Embedded Isolates Predict Fine-Component Loss in Life-Like Cellular Automata

Kunal Bhatia

Independent Researcher, Heidelberg, Germany[a)]

(Dated: 29 April 2026)

Life-like cellular automata contain local structures whose predictive importance depends on which observable describes the system. We identify one such structure: an *embedded isolate*, an alive cell with no orthogonal live neighbour and at least one diagonal live neighbour. Across Conway's Game of Life (GoL) and HighLife, two grid sizes ($L \in \{64, 128\}$), four initial densities ($\rho \in \{0.20, \ldots, 0.35\}$), and seven prediction horizons, the initial count of embedded isolates predicts future loss of fine-scale connected components (mean prestate $R^2 = 0.241$; minimum $R^2 = 0.175$ at $k = 200$; shuffle ≈ 0.0008).

The effect is target-specific: iso_count gives a selective incremental lift for fine-component change while both iso-shuffle and target-shuffle increments are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$, 16 conditions; CV $\lesssim 0.17$).

Mechanistically, embedded isolates with a single diagonal live neighbour have exactly one live Moore-neighbour; under the shared S_{23} survival rule they die at $t+1$ as an exact consequence of the transition table. Multi-diagonal isolates more often survive by merging into a larger component. Both pathways reduce the fine-component count, and a local-window loss feature captures this with $R^2 = 0.538$. After condition standardisation the mechanism transfers across held-out rule, size, and density ($R_z^2 = 0.544$), while raw amplitude is predictable from grid size and density (LOO $R^2 = 0.977$). These results illustrate target-indexed description: embedded isolates are privileged specifically for future fine-component change, not for other summary statistics of the same automaton state.

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response, mechanism transfer, observer-dependent description

[a)]https://orcid.org/0009-0007-4447-6325

I. INTRODUCTION

Conway's Game of Life (GoL)[1] and HighLife are among the most studied cellular automata, valued for their deterministic dynamics, rich emergent structure, and computational tractability. Earlier work established edge-of-chaos properties[2,3] and statistical complexity[4] of Life-like rules, and identified the role of coherent structures[5] and computational irreducibility[6] in deterministic CAs. Local causal-state methods[4,5] identify predictive-equivalence partitions from past/future lightcones; our work instead asks which simple hand-defined structural classes are privileged for a specific coarse-grained target observable. The partial predictability of a chosen observable demonstrated here is consistent with computational irreducibility[6]: we do not claim to predict the full future CA configuration, only one low-dimensional target statistic.

We identify one such privileged class: the count of *embedded isolated cells* (iso_count), alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour. Such cells sit at the topological boundary of 4-connectivity: orthogonally isolated, diagonally tethered, they are the most fragile local connectivity nodes in the grid. When they die or merge, the fine-component count decreases by one. The iso_count statistic measures how many such fragile nodes are present at $t = 0$ and, as this paper shows, is sufficient to predict the aggregate fine-net change at every tested horizon. Earlier observer-scale diagnostics from the same research program motivated this target-specific analysis; they are documented in the public repository.

The main results, established across GoL and HighLife, two grid sizes ($L \in \{64, 128\}$), four density bands ($\rho \in \{0.20, \ldots, 0.35\}$), and seven prediction horizons, are: (1) iso_count predicts future fine-component loss with mean prestate $R^2 = 0.241$, sustained above 0.175 at $k = 200$; (2) the effect is target-specific—incremental lift for density, entropy, and coarse-block targets is near zero, as are both shuffle nulls; (3) the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations, with CV $\lesssim 0.17$; (4) the mechanism is local component-context loss, not simple cell death, and transfers across all leave-one-out axes.

Figure 1 provides the entry point: the structural definition of an embedded isolate and the target-specific ΔR^2 pattern that singles it out.

II. DEFINITIONS AND PROTOCOL

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a). We denote the 4-connected component count at time t as $C(t)$ and define the fine-net change at horizon k as $\Delta C_k = C(t+k) - C(t)$.

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons $\times$ 16 conditions = 112 condition-horizon slope estimates.

The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares with $N_{\text{live}}(0)$ (live-cell count, proportional to density) as a nuisance covariate, so that iso_count is evaluated for its density-orthogonal contribution. R^2 values for mechanism features are cross-validated (5-fold). Slope confidence intervals are obtained by resampling worlds within conditions (500 bootstrap draws), not from parametric OLS standard errors, so that no homoskedasticity assumption is required across the 1000-world batteries. Each initial grid is drawn independently, making worlds the natural bootstrap unit. All results are scoped to these two Life-like rules on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Low-density dynamics ($\rho \lesssim 0.10$), other Life-like rules, and non-random initial conditions are outside the present scope. We do not rerun experiments; all statistics are drawn from the fixed set of CSVs in `outputs/`.

III. TARGET-SPECIFIC SELECTION AND NON-LEAKY PRESTATE

A. Target-specific ΔR^2

Figure 1b shows the selection-principle battery: for each candidate predictor we measure $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$ on six displayed targets (iso_count, coarse block-variance, block entropy, iso-shuffle null, and target-shuffle null; see Appendix A for the residualisation protocol).

iso_count on the fine-net target yields $\Delta R^2 = 0.0012$, a positive incremental lift in both GoL and HighLife screens and in the global pooled screen. Both shuffle nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not an artefact of the density-count correlation. iso_count adds essentially nothing for density, block-variance, block-entropy, or coarse-component targets: the uplift is concentrated on the fine-net target only. This specificity rules out explanations based purely on total alive-cell count or generic spatial statistics.

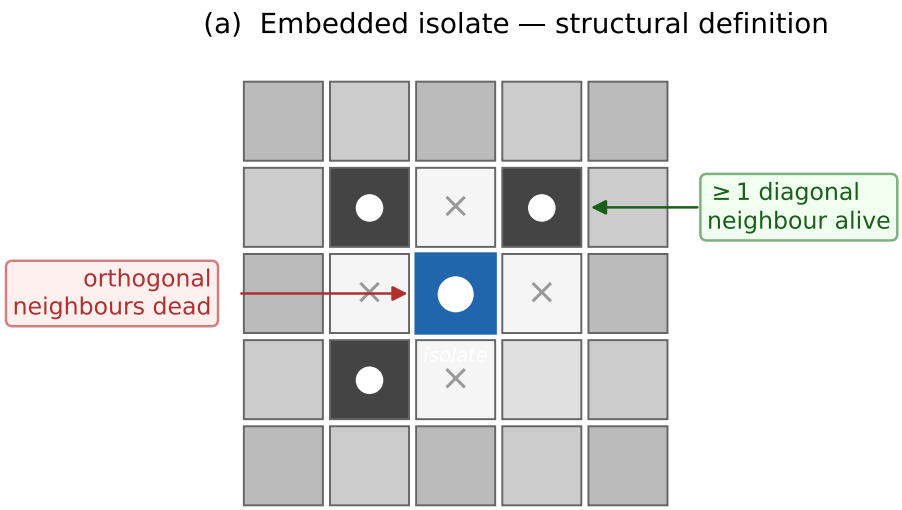
B. Non-leaky prestate recovery

A natural concern is whether iso_count is a proxy for information that requires knowing the one-step fate of individual cells. Figure 2 settles this: using *only* iso_count at $t = 0$—no fate features, no local-window information—the prestate model achieves mean $R^2 = 0.241$ across all horizons $k \in \{1, \ldots, 200\}$ (minimum $R^2 = 0.175$; shuffle ≈ 0.0008). Adding the full diagonal-class breakdown gives no substantial gain. The full multi-horizon response is encoded in the $t = 0$ count alone.

IV. TEMPORAL RESPONSE CURVE

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

All 112 slopes are negative; all 112 bootstrap slope confidence intervals are negative (= 1.00 at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$ (range ≈ 0.09) while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains well above the null at every horizon.



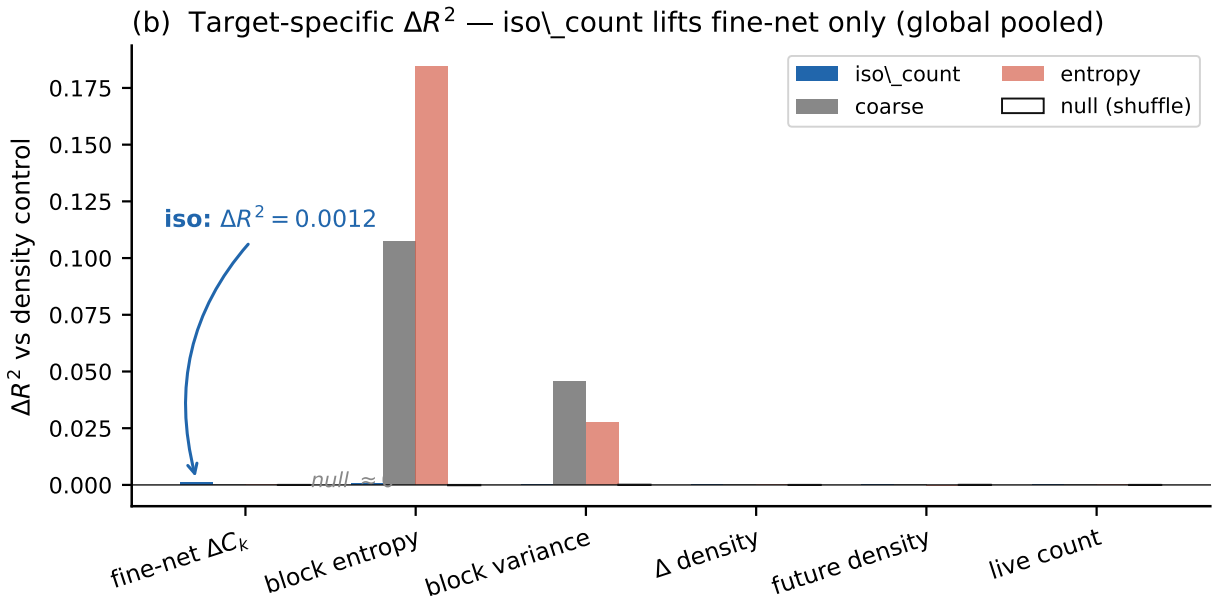


FIG. 1. **Structural definition and target-specific selection.** (a) An embedded isolate: an alive cell (filled circle) with all four orthogonal neighbours dead (×) and at least one diagonal neighbour alive (filled grey). This configuration places the cell at the most fragile locus of the local component topology. (b) ΔR^2 bars for iso_count and comparison predictors (coarse, entropy, null) on six displayed targets (global pooled; `target_delta_components` is numerically identical to fine-net and omitted). iso_count adds a positive incremental $\Delta R^2 = 0.0012$ for fine-net only; both nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not a density artefact.

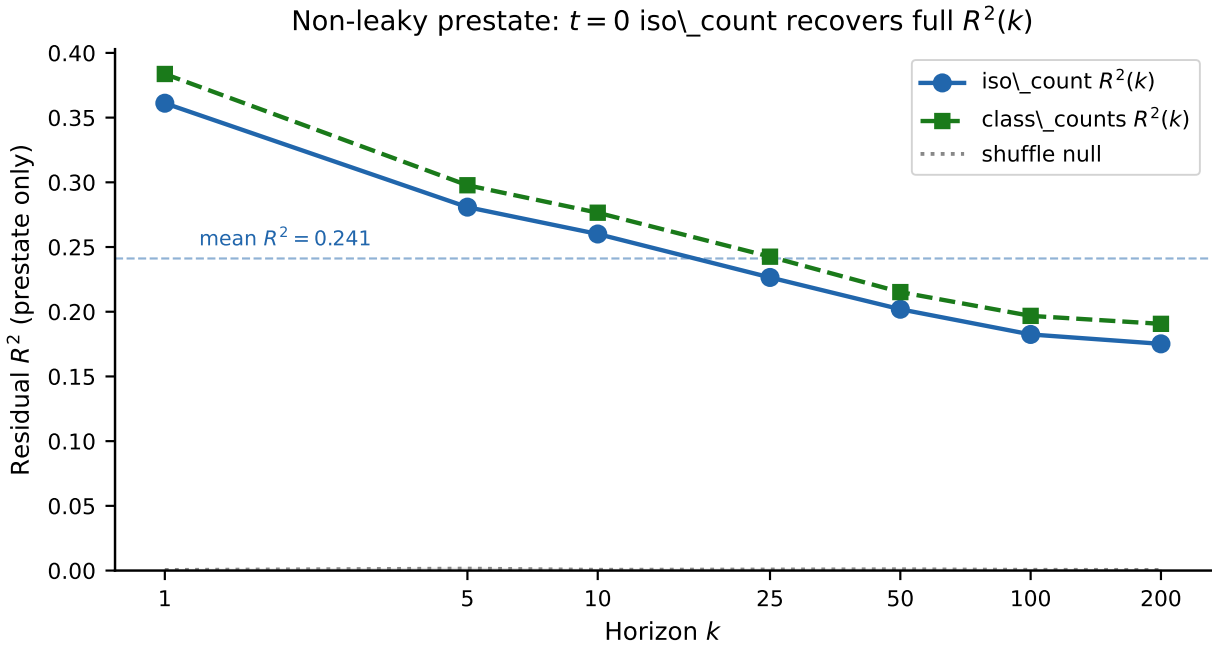


FIG. 2. **Non-leaky prestate recovery.** Using only $t = 0$ iso_count (no fate features, no local-window information), residual $R^2(k)$ stays above 0.175 at every horizon and averages 0.241 (dashed line). Adding class-count breakdown (green) gives marginal improvement. The grey band shows the shuffle null ($R^2 \approx 0.0008$). All temporal prediction is encoded in the $t = 0$ statistic.

Figure 3 shows the slope and R^2 curves. The slope plateau from $k = 5$ to $k = 200$ indicates that the mechanism acts primarily in the first step: once the initial fragile nodes are resolved at $t = 1$, the remaining dynamics do not strongly modulate the iso_count signal. The CV profile confirms robustness: condition-level spread is small relative to the global mean, and GoL and HighLife track each other closely despite different neighbourhood rules.

Figure 4 provides the complete picture: a 16×7 heatmap of all condition-horizon slopes, every cell shaded negative. There are no outlier conditions or horizons.

V. MECHANISM: LOCAL COMPONENT-CONTEXT LOSS

Having established *that* iso_count predicts fine-component change, we turn to *how* this occurs—what actually happens at $t \to t + 1$ when embedded isolates reduce the future component count?

We decompose iso_count into one-step outcome categories: death, survival, survival-

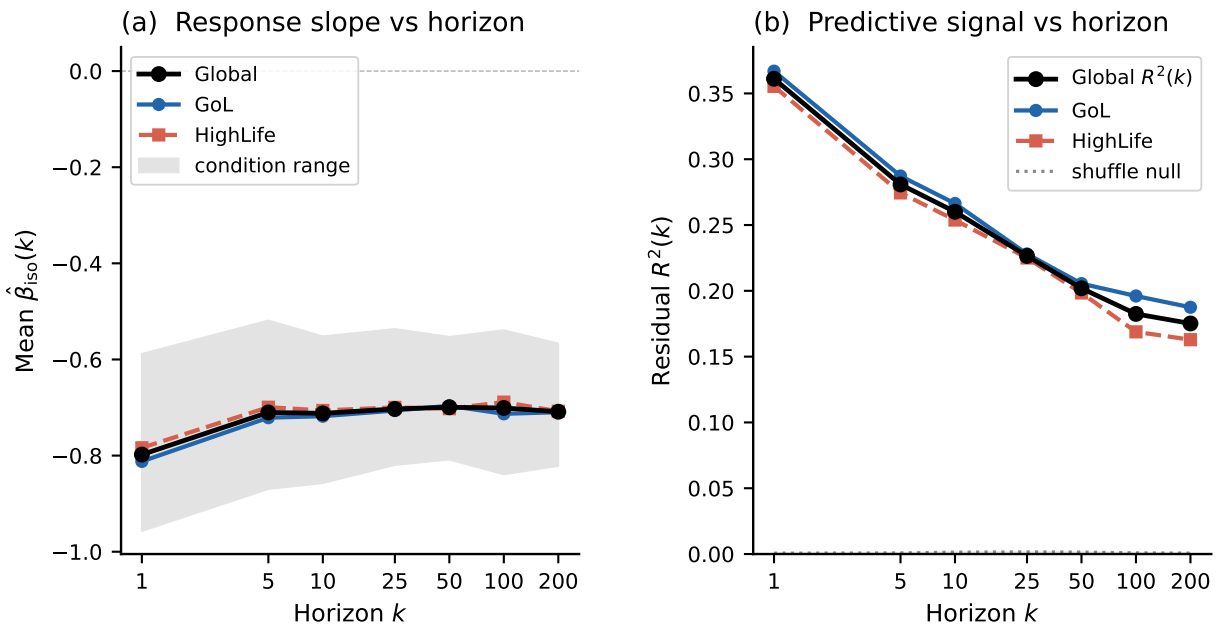


FIG. 3. **Temporal response: slope and predictive signal.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (blue circles), HighLife (orange squares), and global mean (black); shaded region shows condition-level spread. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; grey band marks the shuffle null. Predictive power decays gracefully but remains far above null at all tested horizons.

TABLE I. Temporal response summary (16 conditions, 1000 worlds each; bootstrap CIs, 500 draws). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

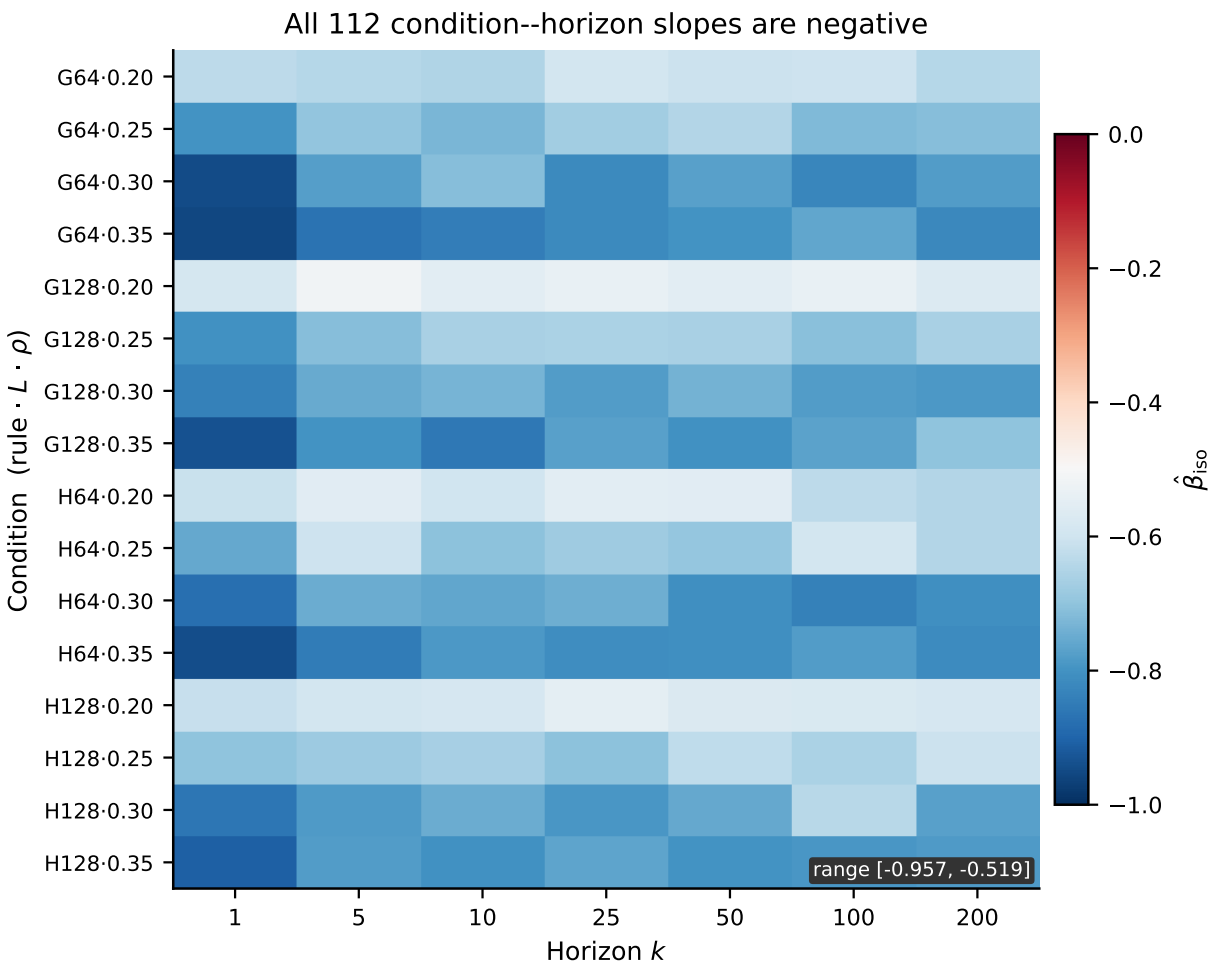


FIG. 4. **All 112 condition–horizon slopes are negative.** Full 16×7 heatmap of $\hat{\beta}_{\text{iso}}$. Rows are conditions (rule $\cdot$ L $\cdot$ ρ), columns are horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$. Every cell is negative, confirming the negative temporal response without exception.

connected (merges with a larger component), or triggering births at orthogonal or diagonal sites. We also compute a *local-window* feature: the total component-count decrease within a 5×5 window centred on each isolate over one step. Cross-validated R^2 values for these features are shown in Fig. 5a.

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net mechanism is *local component-context loss*, not simple cell death. Figure 5b shows the standardised slopes for each fate feature, revealing the partial cancellation between loss and

gain.

Class-level analysis (Fig. 6) reveals a consistent physical story.

a. Exact single-diagonal fate. Embedded isolates in classes 1, 2, 4, or 8 (exactly one diagonal live neighbour, no orthogonal live neighbours) have exactly one live Moore-neighbour at time t. Both GoL (S_{23}) and HighLife (S_{23}) require two or three live Moore-neighbours for cell survival; therefore the focal cell dies at $t+1$. This is an *exact consequence of the transition table*, not an empirical fit. Death fractions of 1.0 for these classes (Fig. 6) are analytically guaranteed. *Multi-diagonal isolates* (classes with two or more diagonal neighbours) depend on the surrounding 5×5 context for births and component merging, and their contributions are measured empirically.

More often they survive-connected: they merge with a larger component and reduce the fine component count by absorbing an independent component. Both pathways lead to the same aggregate outcome—local fine-connectivity loss—which is why the local-window loss feature ($R^2 = 0.538$) dominates over simple death counts ($R^2 = 0.355$ for iso_count alone). The dominant process is the disappearance of fine-net connectivity in the local neighbourhood, aggregated over many isolate classes and many conditions.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

VI. ROBUSTNESS: TRANSFER, AMPLITUDE, AND NULLS

The mechanism analysis tells us what the carrier is. Two robustness questions remain: does it generalise to held-out conditions, and does the scale of the response follow a systematic pattern?

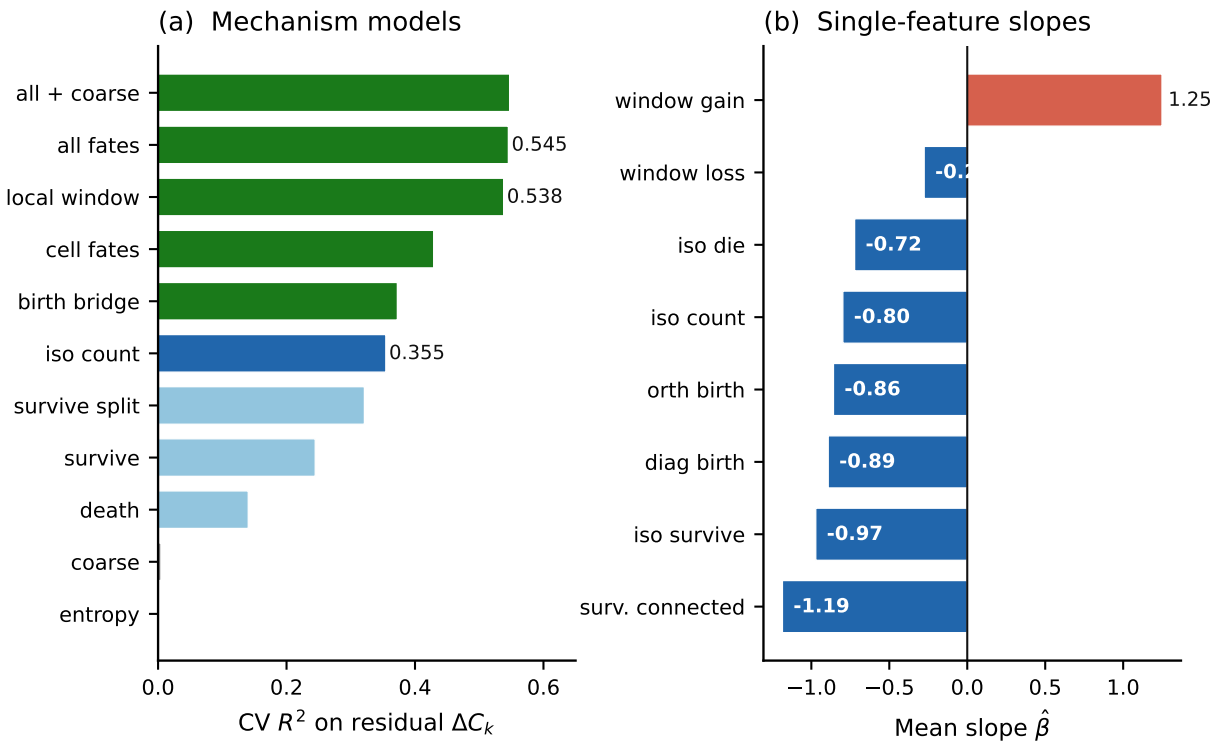


FIG. 5. **Mechanism carrier decomposition.** (a) Cross-validated R^2 for each fate and mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features are negative, gain-related features are positive, revealing partial cancellation.

A. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). All four transfer axes pass (Fig. 7a). For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive $= 1.00$ on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$). The standardised mechanism direction is condition-invariant: whichever leave-one-out split is applied, a model trained on the remaining conditions predicts the held-out condition well.

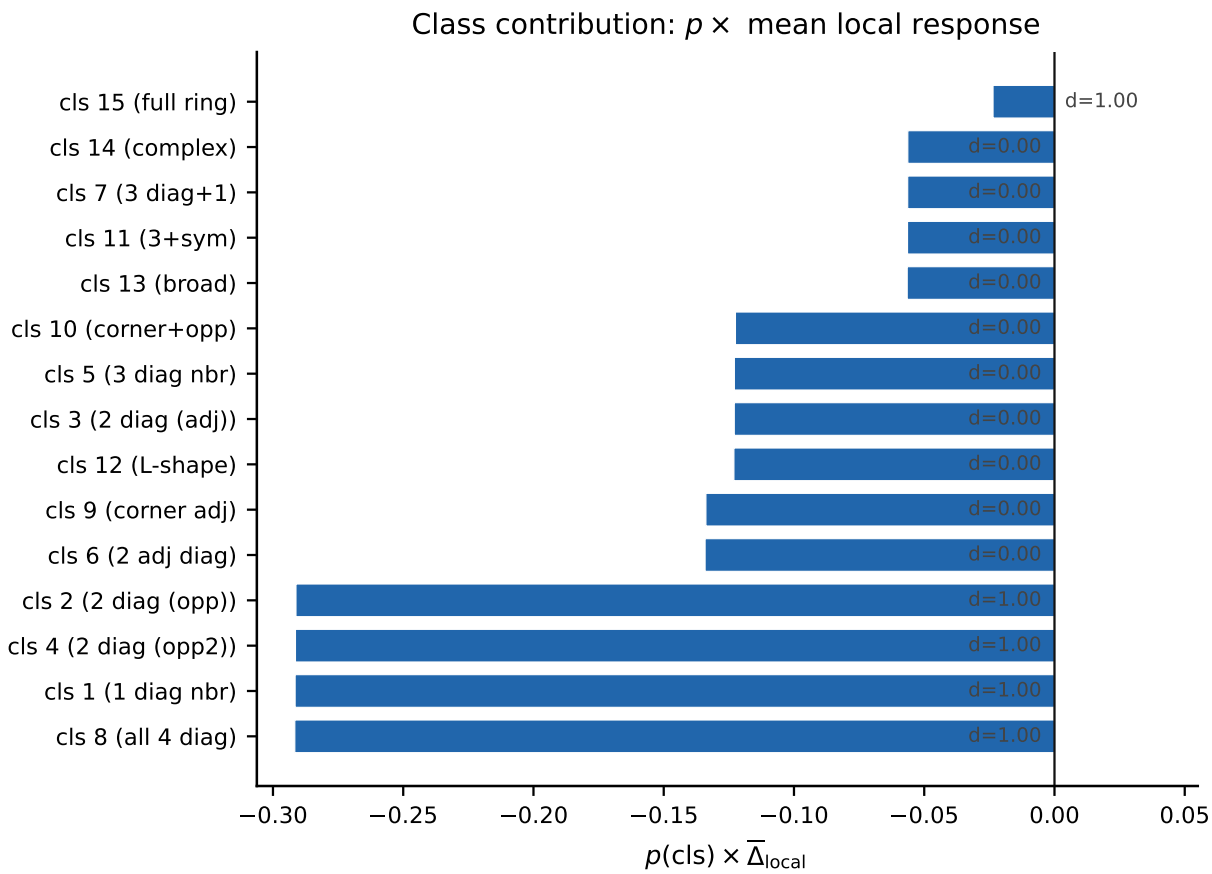


FIG. 6. **Class-level contributions to the mechanism.** Horizontal bars show $p(\text{cls}) \times \overline{\Delta}_{\text{local}}$ for each diagonal-adjacency class. Annotations give the death fraction (d) from GoL transition-class statistics. All classes contribute negatively; single-diagonal-neighbour classes drive the bulk via their high prevalence.

B. Amplitude Law

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV = 0.369) but are predictable from (L, ρ). LOO R^2: full model = 0.977; size+ρ model = 0.970; size-only ≈ 0.802; rule-only $= -0.138$. Grid size L is the dominant predictor of amplitude; rule identity and density alone do not determine it.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L, \rho)\, \hat{\beta}_{\text{iso}}, z\,, \tag{1}$$

where $\hat{\beta}_{\text{iso}}, z \approx -0.50$ is the transferable standardised slope and $A(L, \rho)$ is a condition amplitude predictable with LOO $R^2 \approx 0.977$. The natural explanation is that L sets the

number of components and hence the denominator of the relative change $\Delta C_k/C(0)$, while ρ controls the density of the specific structural class; together they set the amplitude scale that multiplies the universal standardised direction.

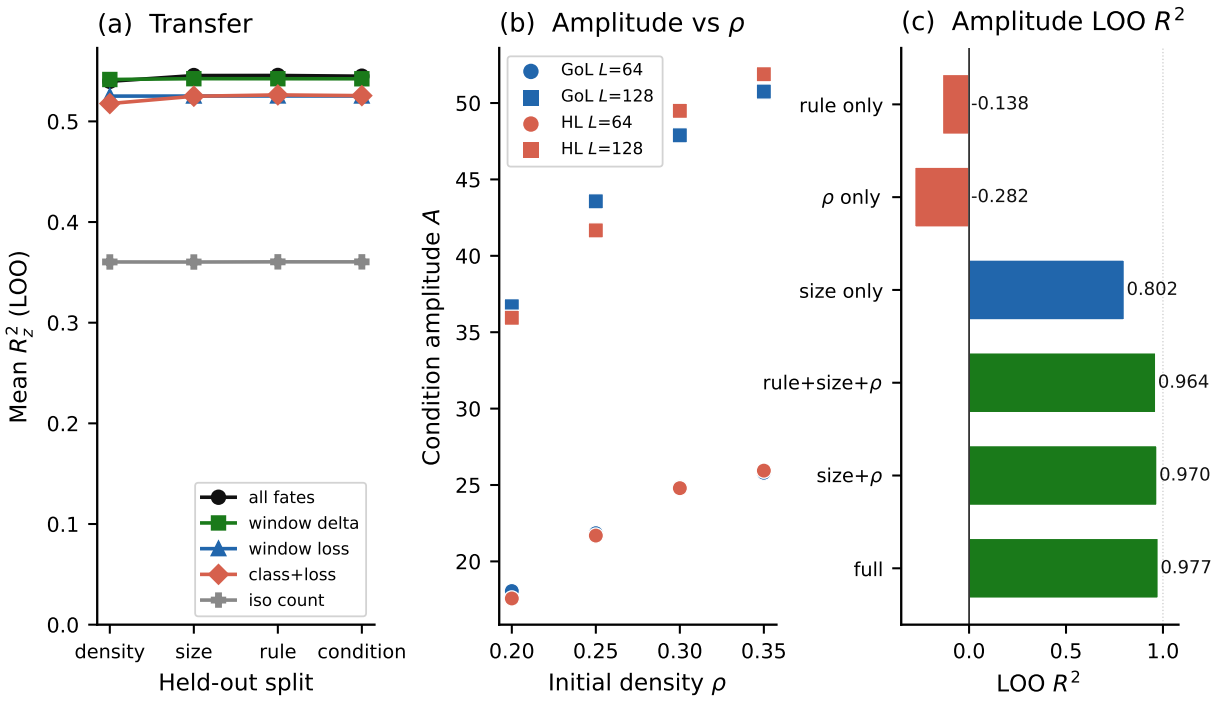


FIG. 7. **Transfer and amplitude calibration.** (a) Standardised transfer R_z^2 for five mechanism models across four leave-one-out axes; all lines stay positive. (b) Condition amplitude versus initial density ρ, coloured by grid size L and rule. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$, size+ρ alone reaches 0.970.

VII. DISCUSSION

Summary. iso_count is a non-leaky, target-specific predictor of fine-net component change (mean prestate $R^2 = 0.241$; shuffle ≈ 0.0008; both shuffle incremental nulls near zero). $\hat{\beta}_{\text{iso}}(k)$ is negative across all 112 condition-horizon combinations, both rules, both grid sizes, and all four density levels. The one-step mechanism is local component-context loss ($R^2 = 0.538$), not simple cell death; single-diagonal isolates die exactly by the S_{23} rule, while multi-diagonal isolates often survive-connected. The standardised direction transfers across

all leave-one-out axes and raw amplitude is calibrated by (L, ρ) with LOO $R^2 = 0.977$.

Common-direction diagnostic. A descriptive check on why a single statistic serves all fine-net horizons is to measure alignment of regression-coefficient vectors in feature space. For each task we fit a residual model and normalise the coefficient vector; pairwise absolute cosines $|\cos\theta_{ij}|$ measure task alignment. Fine-net horizon tasks are near-collinear ($|\cos| = 0.999$, rank-1 energy $= 0.999$): a single feature direction aligns all k-step predictions simultaneously. Heterogeneous targets are not ($|\cos| = 0.427$), and no single predictor adds meaningful ΔR^2 for all of them. Figure 8 shows this contrast. This collinearity is a descriptive geometric property of the regression directions, not a claim about any specific theory of causal emergence or effective coarse-graining.

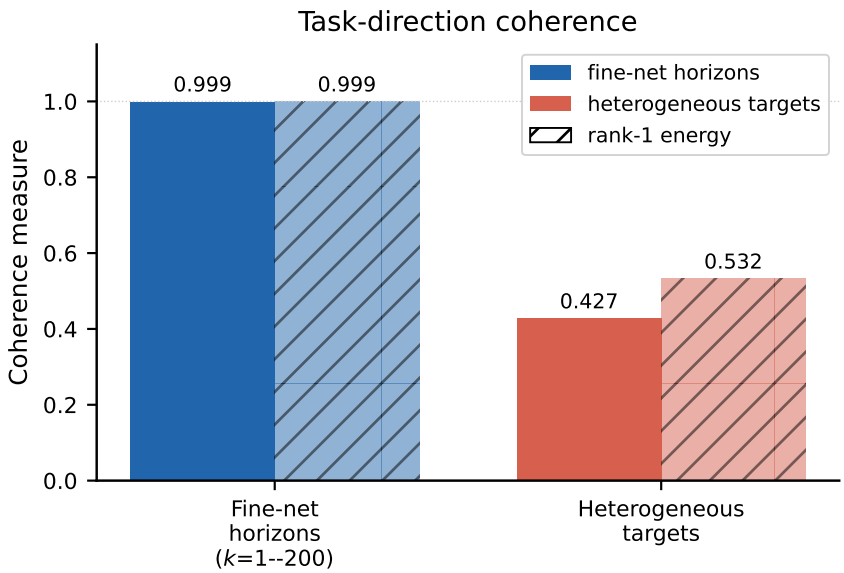


FIG. 8. **Task-direction coherence.** Mean pairwise $|\cos\theta|$ (solid) and rank-1 energy (hatched) for fine-net horizon tasks (blue) and heterogeneous targets (orange). Fine-net horizons are near-collinear ($|\cos| = 0.999$, rank-1 $= 0.999$); heterogeneous targets are dispersed ($|\cos| = 0.427$).

The two-layer structure. The decomposition $\hat{\beta}_{\text{iso}}(\text{cond}) \approx A(L, \rho)\, \hat{\beta}_{\text{iso}}, z$ cleanly separates the universal mechanism from the condition-specific scale. This matters for applications: if a practitioner has access to a new (L, ρ) condition, they can predict the raw slope amplitude with LOO $R^2 \approx 0.977$ from (L, ρ) alone, and they can apply the standardised slope direction directly from the training pool. The two layers are independently portable.

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with

$L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Relation to structural and computational work. Local causal-state methods[4,5] infer predictive-equivalence partitions from joint past/future lightcone statistics. Embedded isolates are a hand-defined structural class within that broader space: they are not claimed to be a full predictive-equivalence partition, but they capture one target-specific component of future fine-connectivity change with a simple geometric definition. The amplitude law does not conflict with coarse-graining impossibility results[6]: we predict one low-dimensional target statistic (ΔC_k), not the full future CA configuration. The partial predictability shown here is consistent with computational irreducibility; it concerns a specific chosen observable, not global state reconstruction.

Embedded isolates as objects. Embedded isolated cells form a well-defined structural class at the topological boundary of connectivity. Their prevalence determines how much the fine-component count is "at risk" at the next step. The stability of $\hat{\beta}_{\text{iso}}(k)$ across horizons suggests that the risk resolves primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result. iso_count is thus a *topological risk indicator* for fine-net change, not a simple density proxy.

VIII. CONCLUSION

The count of embedded isolated cells is the sufficient non-leaky prestate predictor of future fine-component change in Conway's Game of Life and HighLife. The $t = 0$ count alone explains a substantial fraction of residual fine-component change across all tested horizons (mean prestate $R^2 = 0.241$; minimum $R^2 = 0.175$ at $k = 200$; shuffle ≈ 0.0008). The residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations (7 horizons $\times$ 16 conditions; 2 rules $\times$ 2 sizes $\times$ 4 density levels), an empirical regularity that holds without exception across the scoped battery. Both iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The one-step mechanism is local component-context loss, not simple cell death: single-diagonal isolates die exactly

under the shared S_{23} rule; multi-diagonal isolates often survive-connected. This standardised mechanism transfers across all leave-one-out axes and the raw amplitude is predictable from (L, ρ) (LOO $R^2 = 0.977$). The collinearity of fine-net regression directions ($|\cos| = 0.999$) provides a compact geometric diagnostic of why one prestate statistic serves all tested horizons simultaneously.

ACKNOWLEDGMENTS

The author thanks the open-source communities behind NumPy, SciPy, scikit-learn, pandas, Matplotlib, and tqdm.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts of interest to disclose.

DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`. The repository includes an automated regression suite (93 tests) covering the selection, horizon-response, mechanism, transfer, amplitude-law, and artifact-generation pipelines; all tests pass for the submitted build.

Appendix A: Regression Details and Null Definitions

Residualisation protocol

For each prediction world, the density nuisance covariate is $N_{\text{live}}(0)$, the total live-cell count at $t = 0$ (proportional to ρ up to Bernoulli sampling variance). All regression models include $N_{\text{live}}(0)$ as a linear covariate before adding iso_count or any fate feature. $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$. The residual slope $\hat{\beta}_{\text{iso}}$ is the OLS coefficient on iso_count in this density-residualised fit.

Null definitions

Two nulls are used: (i) *iso-shuffle null*: iso_count labels are permuted uniformly at random within each condition; ΔR^2 and slope are recomputed. (ii) *target-shuffle null*: the target ΔC_k is permuted; ΔR^2 is recomputed with the original iso_count. Both converge near zero ($\Delta R^2 < 0.001$), confirming the signal depends on the joint alignment of iso_count with ΔC_k and is not a density or permutation artefact.

Appendix B: Transition-Class Coding

Diagonal-adjacency class is an integer in $[1, 15]$ indexing the bit pattern of the four diagonal neighbours of an embedded isolate (NW, NE, SW, SE), with class c equal to the binary value $NW \cdot 1 + NE \cdot 2 + SW \cdot 4 + SE \cdot 8$. Class 15 (all four diagonal neighbours alive) is rare in this battery ($p \approx 0.009$, $n = 80{,}015$ instances) but contributes a negative local response ($\bar{\Delta} \approx -2.52$, death fraction $= 1.0$) comparable to other multi-diagonal classes; it is not excluded from the analysis. Classes 1, 2, 4, 8 are the *single-diagonal* classes; classes 3, 5, 6, 9, 10, 12 are the *two-diagonal* classes; classes 7, 11, 13, 14 are the *three-diagonal* classes; class 15 is the *four-diagonal* class.

REFERENCES

[1]M. Gardner, "Mathematical games: The fantastic combinations of John Conway's new solitaire game "life"," Scientific American **223**, 120–123 (1970).

[2]C. G. Langton, "Computation at the edge of chaos: Phase transitions and emergent computation," Physica D: Nonlinear Phenomena **42**, 12–37 (1990).

[3]S. Wolfram, "Universality and complexity in cellular automata," Physica D: Nonlinear Phenomena **10**, 1–35 (1984).

[4]J. P. Crutchfield and K. Young, "Inferring statistical complexity," Physical Review Letters **63**, 105–108 (1989).

[5]A. Rupe and J. P. Crutchfield, "Local causal states and discrete coherent structures," Chaos **28**, 075312 (2018).

[6]N. Israeli and N. Goldenfeld, "Computational irreducibility and the predictability of complex physical systems," Physical Review Letters **92**, 074105 (2004).